



Fig. 6: Jumper wires connected to the screen

need to decide its features and understand how it would work. As soon as the robot is switched on, it gets ready to accept the voice command and indicates its readiness through the glowing LED indicator on top of its head. The LED glows when the robot is ready and listening; it switches off while processing the voice command.

When the robot is asked to introduce itself and to solve a sudoku, it switches on camera. The robot looks for the puzzle and on finding it solves it automatically. It automatically creates the code for all these functions. It is that simple!

Assembly

Now, all you have to do is to assemble the components and get the robot

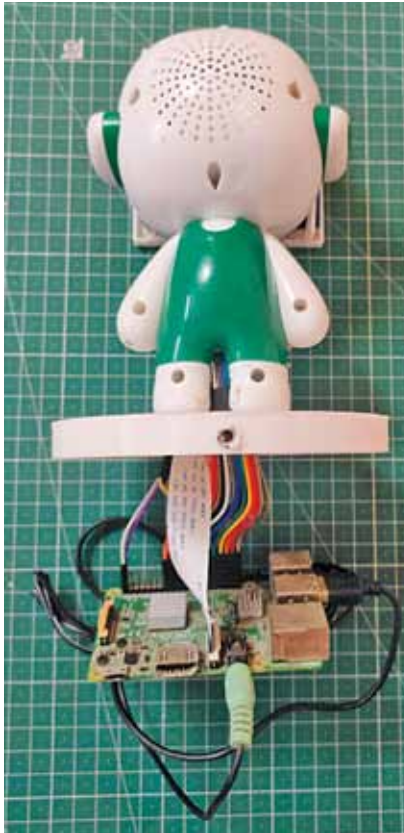


Fig. 7: Raspberry Pi connected to other parts in the robot

ready for programming. First, insert the SD loaded with latest Raspbian OS and connect the camera module to Raspberry Pi camera port and the speaker to Raspberry Pi Aux port.

Connect the LED to any of the GPIO pins of Raspberry Pi (connected with the GPIO pin 17 in prototype, as shown in Fig. 3). Next, connect the female-to-female jumper wire connector to Raspberry Pi and the LCD display, aligning the pin from first pin of Raspberry Pi to first pin of LCD display, and so on, as shown in Fig. 4.

Setting up Raspberry Pi for coding

For programming, connect the Raspberry Pi with a large display through HDMI, or simply use SSH. After getting access to the Pi, install the required drivers for the Raspberry Pi screen (details available in the



Fig. 8: Final looks of the prototype robot

touch-screen's manual). Then enable the camera using raspi-config. Next, set Audio Output to 3.5mm Jack, using raspi-config or Settings.

Now connect the Raspberry Pi to Wi-Fi so that it can be controlled without the large display also. After this, set up the mic and test it. In this step, editing of the file `/home/pi/.asoundrc` would be required.

After doing the basic setup in Raspberry Pi, fit the components in a toy robot body. You can design your own customised robot body and 3D print it or you can buy and use a readymade robot's body of your choice. For the prototype, an old toy robot's body was used and the components fitted as shown in Fig. 5 through Fig. 8. The finished robot can be seen in Fig. 8.

Programming the robot

Python3 programming language may be used to program the robot. To make the coding simple and straightforward, all actions of the robot—like speech recognition, face animation, and sudoku solving—are performed using a single Python